

1. Main Sources Studied

We studied 12 sources ten peer-reviewed papers, one Microsoft/WWF technical platform,one internal design note.

Source	What it contributed to DRISHTI	Importance
SW-Net-Sensors (2026)	Used its shipwreck results to compare our model.It also showed that even human experts can disagree when labelling sonar images.	High
DFSE-YOLO-Scientific Reports (2026)	Used its YOLOv8 results as a detection baseline and used its comparison with other models to support our model choice.	High
Transformer-YOLOv5-Remote Sensing (2021)	Used its method for finding the real-world location of detected objects from sonar data.	High
GhostVision-JMSE(2026)	Helped us understand why precision and recall are important for ghost-net detections.We also used its idea of checking detections over time	High
GhostNetZero-Microsoft AI for Good/WWF	Provided reference results for ghost-net detection and showed the difficulty of using data from one region in another.	Targeted
Du et ai. (2023)-Remote sensing	Showed that some sonar datasets are not always better for pretraining than general image datasets.	Targeted
Qin et ai. (2021)-IEEE Access	Supported the use of transfer learning when only small amount of sonar training data is available.	Targeted
Du et al.(2025)-JMSE	Showed that having too many examples of one class can affect the detection of other classes.	Targeted
Jiang et al.(2024)-JMSE	Introduced WIoU v3 which may consider later for difficult or irregular object shapes.	Minor
Yu et al.-Debris-flow deposits	Supported the idea of using	Minor

	multiple stages of transfer learning for small sonar datasets.	
Cuff et al. (2023)-Fisheries Ecology	Helped provide a scientific basis for deciding how many consecutive sonar frames should be checked together	Minor
SIH 26057-Internal Note	Used as a checklist for GPS, coordinate systems, and sonar-ping matching. It also ensures that we do not create false GPS	Minor

Complete Literature

The reviewed studies show that deep-learning-based object detection can be effective for sonar imagery, but performance depends strongly on the availability and quality of labelled data, the characteristics of the sonar images, and the similarity between training and deployment conditions. The literature also indicates the usefulness of transfer learning when task-specific datasets are limited.

For DRISHTI, these findings support the use of a pretrained YOLO-based detector followed by fine-tuning on the available sonar dataset. The reviewed work also highlights the importance of preprocessing, hard-negative examples, appropriate evaluation, and testing on data that represents realistic deployment conditions. These observations guided the dataset preparation, model selection, and evaluation strategy described in the following sections.

2.The architecture families

2.1 Shipwrecks:-Shipwrecks are usually large objects with strong acoustic signatures. Researchers have used: CNN-based models, U-Net, YOLO, segmentation models, Improved object detection networks. YOLO-based detection is useful when we need bounding boxes and faster inference. This was one reason we selected YOLOv8s for DRISHTI.

2.2 Ghost Nets:-Ghost nets are usually much harder to detect. They can appear as thin or broken structures in sonar images. Research has therefore used segmentation-based approaches such as DeepLabV3 and other specialised networks. Segmentation can be useful because it identifies pixels belonging to the target rather than only drawing a bounding box. However, segmentation models can require more computation. Since DRISHTI is intended to eventually work on an edge computer attached to an underwater vehicle, we had to consider both accuracy and computational cost.

2.3 Pipes and Cylinders:-Pipes and cylinders have more regular shapes than ghost nets. Some research uses geometric information along with deep learning. The literature also discusses physical sonar models such as Lambertian reflection models. These models can help explain why an object becomes brighter or darker depending on its position, angle, and distance from the sonar. For DRISHTI, this supports the use of geometry and acoustic-shadow information along with the detector output.

Why We Used One Detector

Based on the reviewed literature and the project requirements, DRISHTI uses a single multi-class detector to identify the target categories within the same inference pipeline. The detailed justification for this model-selection decision, including the comparison with separate specialist detectors, is discussed in File 4.

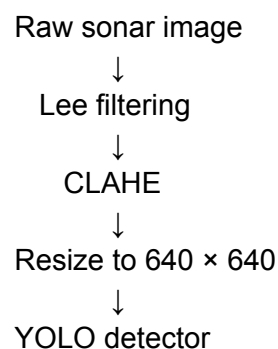
3. Sonar Image Denoising

Side-scan sonar images contain different types of noise. Because of this, several research studies have explored deep learning-based denoising. Methods such as self-supervised denoising can be useful because clean sonar reference images are usually not available. Examples discussed in the literature include: Neighbor2Neighbor, Blind2Unblind and DSPNet.

Some research reports improvements in downstream object detection after denoising. This made deep denoising an interesting option for DRISHTI. However, we did not automatically assume that a more complex denoising model would improve our system. We tested preprocessing on our own data. Our comparison did not show a major improvement in detector accuracy. Therefore, we decided to keep a simpler preprocessing pipeline using Lee filtering and CLAHE. This is an important part of our literature review because it shows that we did not simply choose the most complicated method available. We tested the idea and selected the method that better suited our current project.

Why Lee Filter and CLAHE

The Lee filter is a traditional method used for reducing speckle noise. It is useful because it tries to reduce noise while preserving important edges. This is important in sonar images because excessive smoothing can remove the boundaries of an underwater object. CLAHE stands for Contrast Limited Adaptive Histogram Equalization. It improves local contrast in an image. This can make bright and dark regions easier to distinguish. For DRISHTI, we use:



The main advantage of this approach is simplicity. It does not require another large neural network before the detector. It is also easier to inspect and debug.

4. Important Published Benchmarks

Published result	Source	Meaning
0.716 mAP@50	Vanilla YOLOv8 on AI4Shipwrecks	Shipwreck detection reference
0.755 mAP@50	DFSE-YOLO	Improved shipwreck detection result
90–94% detection rate	GhostNetZero	Ghost-net detection in its own survey regions
50–60% agreement	SW-Net / AI4Shipwrecks	Approximate human agreement for wreck boundaries

These numbers need to be interpreted carefully.

The 0.716 and 0.755 results are based on shipwreck detection.

DRISHTI has four product classes.

Therefore, our 0.641 multi-class mAP@50 should not be directly described as worse than these results.

Similarly, the 90–94% GhostNetZero result should not be compared directly with our ghost-net result because the datasets are different and our ghost-net data is synthetic.

The 50–60% human agreement result is useful because it shows that sonar annotation itself has uncertainty.

Normal seabed images help reduce false detections.

5. How the Literature Influenced DRISHTI

The literature was directly connected to our design decisions.

Literature finding	DRISHTI decision
YOLO works well for sonar detection	Use YOLOv8s
Shipwreck data is available through AI4Shipwrecks	Use it for shipwreck training
Ghost nets are difficult to detect	Keep ghost-net handling separate and clearly report synthetic-data limitations
Synthetic data can be risky	Use it only where real data is unavailable
Background is important	Include hard-negative background tiles
Denoising can help but is not guaranteed	Test preprocessing instead of assuming improvement
Deep denoisers add complexity	Use Lee + CLAHE for the current pipeline
Different classes behave differently	Use per-class confidence calibration
Human review is useful	Include operator review in the workflow
Cross-region performance can decrease	Plan for future regional fine-tuning